# BEng (Hons) Software Engineering

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## Introduction

The personal task manager (GUI) features a user-friendly interface built using Tkinter designed to help users to manage their personal tasks effectively. All task data is stored in a tasks.json file, with each task containing a title, description, priority level and due date. At the top of the window users can apply filters to sort tasks by name, priority or due date. The center of the application showcases a table (Treeview) that displays each task's details. Additionally, users can sort the tasks by clicking on column headers for better organization

## Purpose of using the GUI

The purpose of the using the GUI interface enables users to view tasks from a JSON file in a tabular format, filter tasks by name, priority and due date and sort tasks by column headers using by name, priority and due date

## GUI Interface Buttons

1) Title Bar

- Displays the application title as Personal Task Manager

2) Search and Filter Section

- Type any part of a task name to filter
- Choose High, Medium, or Low
- Filter tasks by due date YYY-MM-DD format
- Filter task list based on selected criteria

3) Task Display Table

Treeview widget will display columns Task name, Description, Priority, Due date and tasks are shown in a structured table and clickable column header for sorting

4) Sort Buttons

- Sort by Name
- Sort by Priority
- Sort by Due Date

## Test Cases and Results

Test Case 01 – Sort by Name

*Table 1 - Sort by Name*

| Test Case | Input | Expected Outcomes | Actual Outcomes | Result |
|---|---|---|---|---|
| Sort by Name | Click "Sort by Name" | Tasks sorted alphabetically by name | Tasks sorted correctly in alphabetically order | Pass |

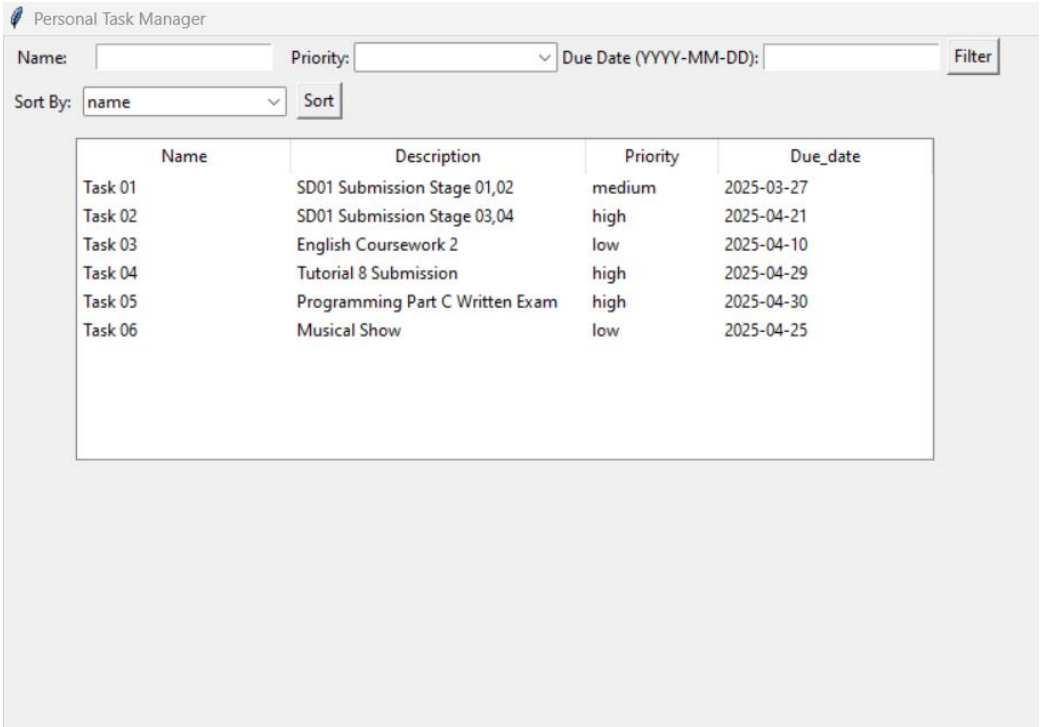


*Figure 1 - Sort by Name (GUI)*

Test Case 02 – Sort by Due Date

*Table 2 - Sort by Due Date*

| **Test Case** | **Input** | **Expected Outcomes** | **Actual Outcomes** | **Result** |
|---|---|---|---|---|
| Sort by Due Date | Click "Sort by Due Date" | Tasks sorted by ascending due date | Tasks appear in correct date order | Pass |

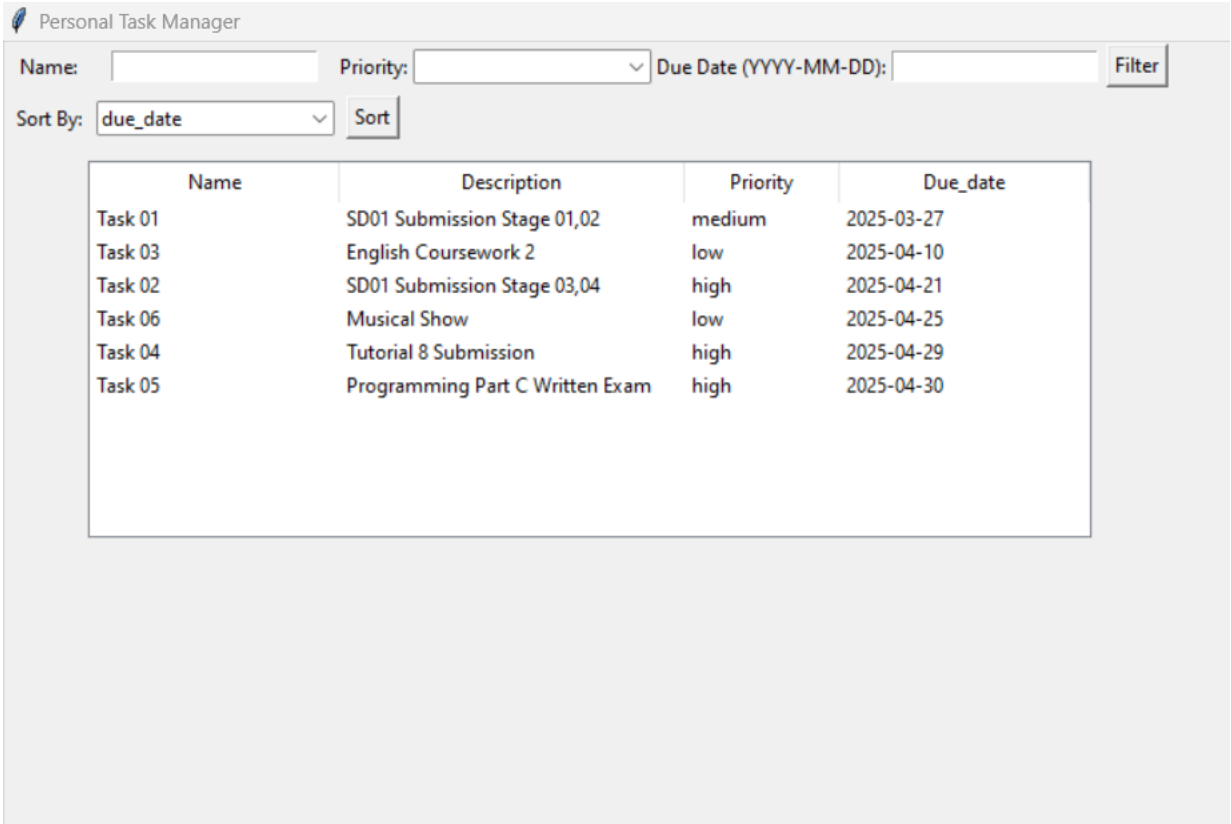


*Figure 2 - Sort by Due Date (GUI)*

Test Case 03 – Enter Task Name in Name Filter

*Table 3 - Enter Task Name in Name Filter*

| Test Case | Input | Expected Outcomes | Actual Outcomes | Result |
|---|---|---|---|---|
| Enter Task Name in Name Filter | Enter Task 02 in Name Filter | Only task with "Task 02" appears | Task filtered correctly | Pass |

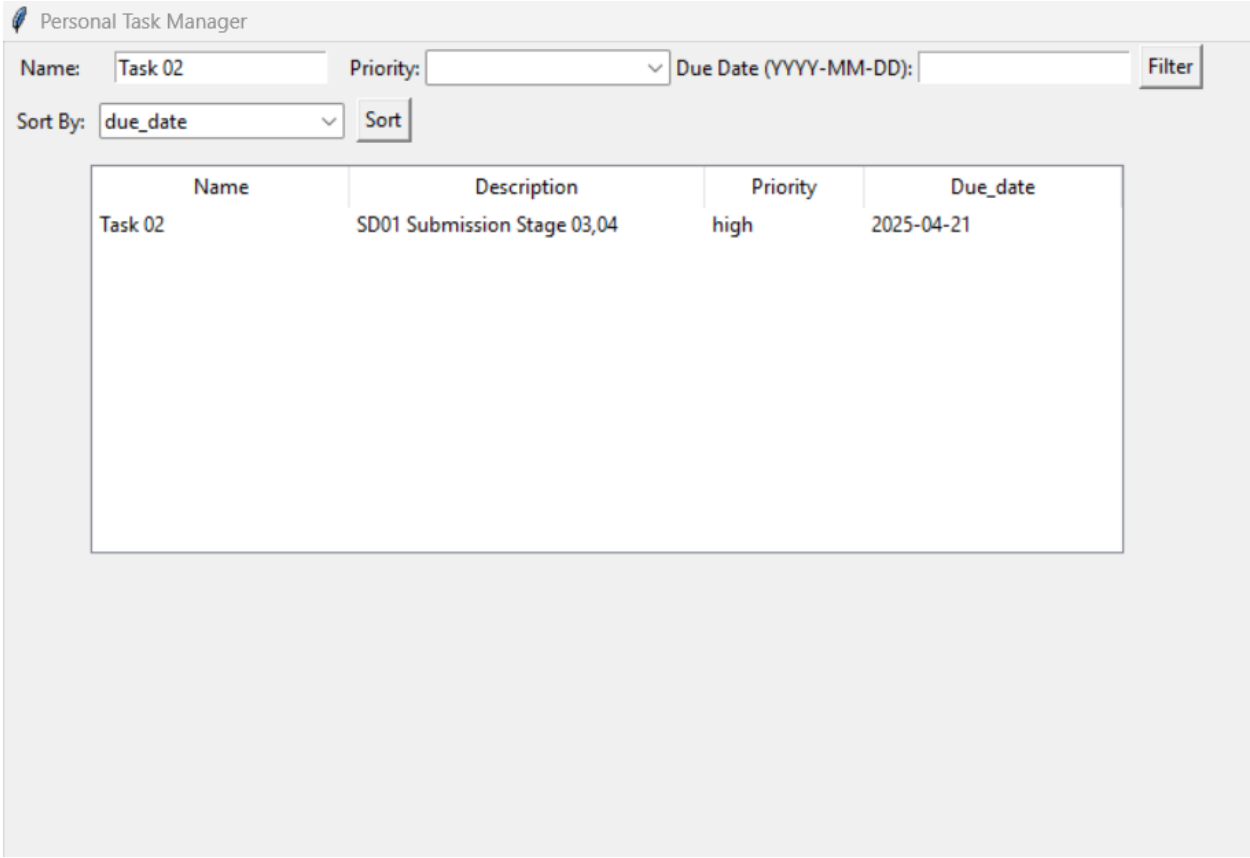


*Figure 3 - Enter Task Name in Name Filter (GUI)*

Test Case 04 – User to enter Due Date filter

*Table 4 - User to enter Due Date filter*

| Test Case | Input | Expected Outcomes | Actual Outcomes | Result |
|---|---|---|---|---|
| User to enter Due Date filter | Enter 2025-04-21 in Due date filter | Only tasks due on that date are shown | Task list correctly filtered | Pass |

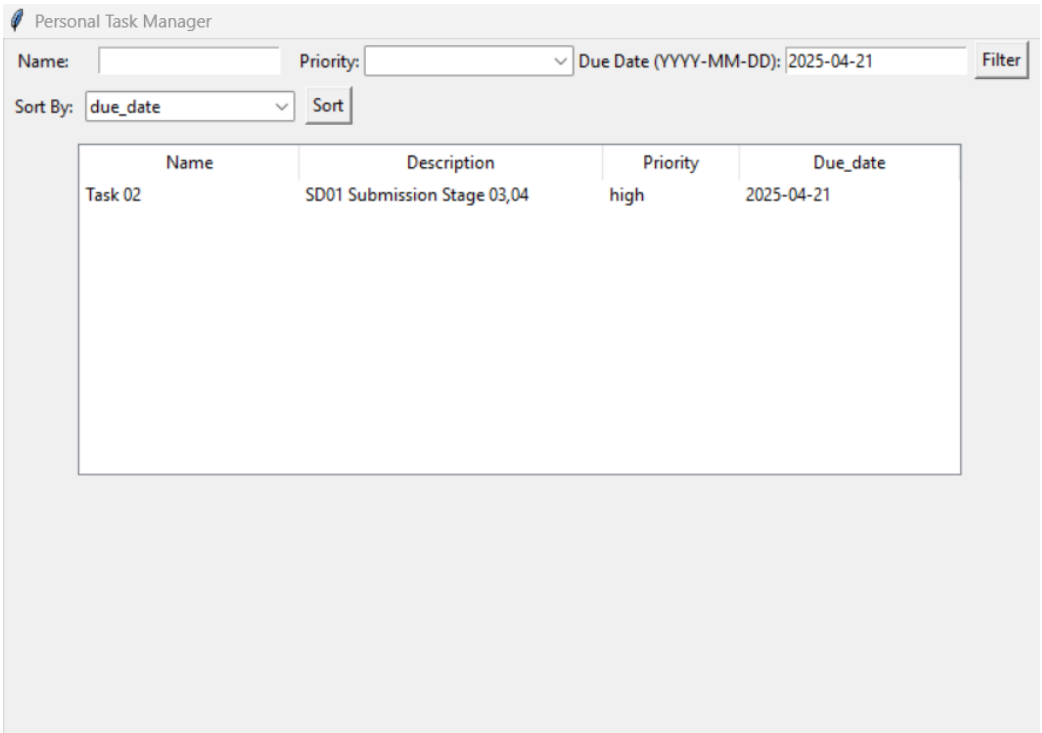


*Figure 4 - User to enter Due Date filter (GUI)*

Test Case 05 – Enter column header to select priority

*Table 5 - Enter column header to select priority*

| Test Case | Input | Expected Outcomes | Actual Outcomes | Result |
|---|---|---|---|---|
| Enter column header to select priority | Click column header "priority" | Tasks sorted by priority | Tasks recorded correctly | Pass |

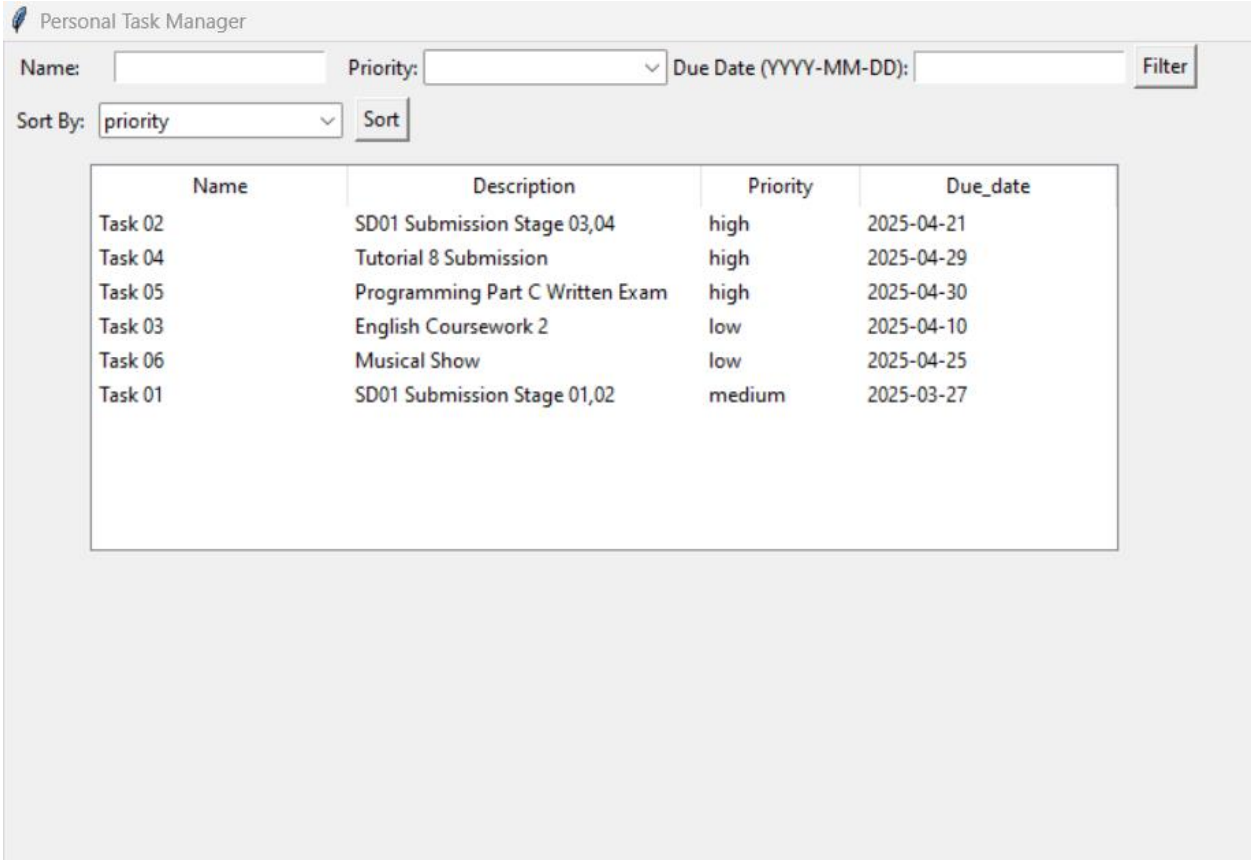


*Figure 5 - Enter column header to select priority (GUI)*

Test Case 06 – Check "High" from priority

*Table 6 - Check "High" from priority*

| Test Case | Input | Expected Outcomes | Actual Outcomes | Result |
|---|---|---|---|---|
| Check "High" from priority | Click "High" from priority dropdown | Only "High" priority tasks are shown | Filter applied successfully | Pass |

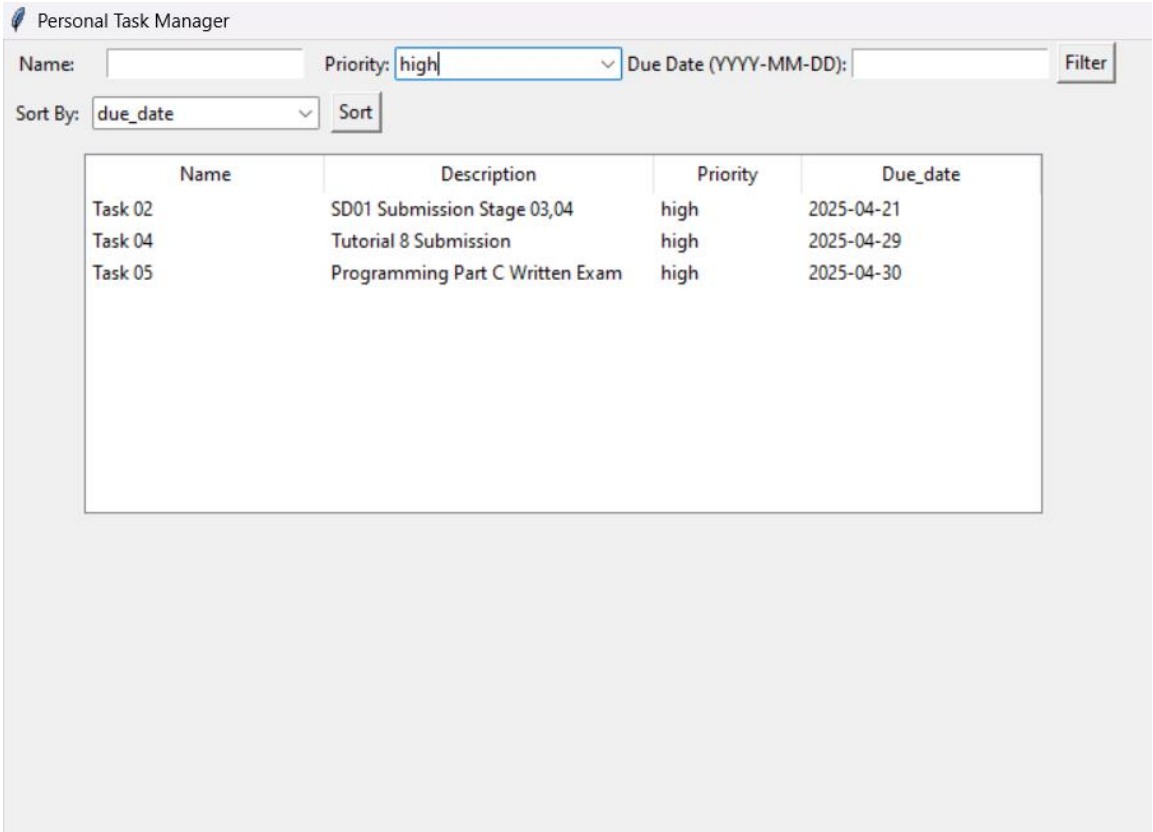


*Figure 6 - Check "High" from priority (GUI)*

# Conclusion

In conclusion, the GUI- based personal task manager successfully provides an organized and interactive platform for managing personal tasks. With features such as filtering, sorting and clear display of task details, users can effectively handle tasks. The use of Tkinter along with JSON storage ensures a smooth and user-friendly experience. Making task management and effective